Ki-Ha Kim (김기하)

HPC / GPU Computing · CUDA C/C++ · Parallel Numerical Solvers

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SUMMARY

Computational scientist specializing in GPU-accelerated CFD solvers and parallel numerical libraries on NVIDIA A100/H100/H200/GH200 — PhD in computational science & engineering, with 5+ years across national and industrial HPC labs. First / co-first author of multi-GPU **tridiagonal and block-tridiagonal** solver libraries with **register-resident CUDA kernels** and **communication-avoiding** algorithms — **up to 42.9× over CPU on H100** and **6.4–6.8× over standard all-to-all on H200** — scaling production simulations to **128 GPUs / 8.5-billion degrees of freedom** and **262,144 CPU cores**, and generating **multi-billion-grid turbulent-flow DNS (up to ~30 billion points)**. Strong **linear-algebra and numerical-methods** foundation; experienced delivering GPU solutions into partners' large production codes alongside domain scientists. 9 papers (7 first/co-first author; CPC / JCP / JFM).

TECHNICAL SKILLS

- **GPU / Parallel:** CUDA C/C++, CUDA Fortran, multi-GPU (NVLink / NVSwitch), CUDA-aware MPI, MPI; register-resident kernels, Nsight Systems profiling
- **NVIDIA Libraries:** cuSPARSE, cuBLAS, cuFFT, AmgX
- **Numerical / HPC:** tridiagonal & block-tridiagonal solvers, Krylov (BiCGStab/CG) with ILU/AMG preconditioners, spectral-element method, DNS/LES; strong/weak scaling to 262K cores
- **Languages:** C++ (CUDA C/C++), Fortran 90/95, Python, C
- **Platforms:** NVIDIA A100/H100/H200/GH200; Linux, SLURM, Git

EXPERIENCE

KISTI — Senior Researcher · Korea Institute of Science and Technology Information · 2024–Present

- **PaScaL_BTDMA — multi-GPU block-tridiagonal solver library (first author): 6.4–6.8× over standard all-to-all** and **91–97% weak-scaling on 8× H200**; strong-scaled to **64 GH200 (1024^3)** on the KISTI-6 "Hangang" system.
- Cut inter-process communication to two boundary block-rows (independent of system length) in a single fused **register-resident CUDA kernel** (0 register spills); the all-to-all bottleneck was isolated with **Nsight Systems**.
- **PaScaL_TDMA 2.1 — register-resident multi-GPU tridiagonal solver (first author): 21.7× / 42.9× on A100 / H100**; consolidated the reduced-system exchange into a **single MPI_Alltoall** and cut global-memory loads (9→4 per iteration), keeping **communication below 10%** of runtime.
- **GPU sparse linear solver for the CUPID reactor code:** built a CSR `Ax=b` solver (BiCGStab, ILU, AMG) on **cuSPARSE / cuBLAS / AmgX — 6.7× over a 64-core CPU** on a 1.48M-cell pressure system (>80% of solver runtime) — and delivered it into the production Fortran code via a **C ABI**.

Samsung SAIT — Senior Researcher · Samsung Advanced Institute of Technology · 2022–2024

- **GPU-LICA — multi-GPU LES solver for semiconductor-fab cleanroom flows:** ran an **8.5-billion-DoF (4096×960×2176) cleanroom LES on 128 A100 GPUs at 85.8% strong-scaling efficiency**, completing one full air-circulation in **about one day**.
- **Optimized multi-GPU communication:** replaced a collective `MPI_Alltoall` with a topology-aware point-to-point exchange, cutting a device-to-device-copy bottleneck from **~35 ms to ~5.3 ms (≈6.6×)**; implemented a **cuFFT-based DCT pressure-Poisson solver** confined to NVLink-connected intra-node GPUs.
- Accelerated the conjugate-gradient solver in charge equilibration (Qeq) of **LAMMPS-ReaxFF** via non-blocking ghost-atom communication.

KIAPS — Senior Researcher · Korea Institute of Atmospheric Prediction Systems · 2020–2022

- Built a **global high-resolution weather-prediction code**: spectral-element method for the non-hydrostatic equations; multi-resolution grids around the Korean Peninsula.

Yonsei University — Postdoctoral Researcher · 2020

SELECTED PUBLICATIONS

9 papers in HPC and computational fluid dynamics — 7 first / co-first author (incl. 1 manuscript), 2 co-author. Full list on Google Scholar.

First / co-first author

1. **K.-H. Kim** et al. *PaScaL_BTDMA: A scalable parallel library for batched block-tridiagonal solvers on distributed multi-GPU systems.* **Computer Physics Communications** (under review, 2026).
2. **K.-H. Kim** et al. *PaScaL_TDMA 2.1: A register-resident multi-GPU tridiagonal matrix solver with optimized communication for large-scale CFD.* **Computer Physics Communications** 323:110120 (2026).
3. **K.-H. Kim**, J.-H. Kang, X. Pan, J.-I. Choi. *PaScaL_TCS: A versatile solver for large-scale turbulent convective heat transfer.* **Computer Physics Communications** 290:108779 (2023). — 131,072 cores; DNS up to ~30 billion grid points.
4. **K.-H. Kim**, J.-H. Kang, X. Pan, J.-I. Choi. *PaScaL_TDMA: A library of parallel and scalable solvers for massive tridiagonal systems.* **Computer Physics Communications** 260:107722 (2021). — scaling to 262,144 cores.
5. **K.-H. Kim**, J. Ahn, J.-I. Choi. *Mean thermal energy balance analysis in differentially heated vertical channel flows.* **Physics of Fluids** 33:065120 (2021) — **Editor's Pick**; 3.4-billion-grid DNS of buoyancy-driven wall turbulence.
6. J. Ahn, **K.-H. Kim** *(co-first author)*, X. Pan, J.-I. Choi. *Contribution of Reynolds shear stress to near-wall turbulence in Rayleigh–Bénard convection.* **Int. J. Heat and Mass Transfer** 181:121873 (2021). — multi-billion-grid RBC DNS (up to ~24 billion points).
7. **K.-H. Kim**, J.-I. Choi. *Lock-in regions of laminar flows over a streamwise oscillating circular cylinder.* **Journal of Fluid Mechanics** 855:1060–1087 (2018).

Co-author — non-Oberbeck–Boussinesq natural convection (J. Comput. Phys. 463:111238, 2022); efficient monolithic projection for natural convection (Int. J. Heat Mass Transf. 144:118677, 2019).

PROJECTS / OPEN SOURCE

- **PaScaL_BTDMA** — multi-GPU block-tridiagonal solver (H200/GH200); CUDA Fortran + **CUDA C++**. github.com/k-kiha/PaScaL_BTDMA
- **PaScaL_TDMAcuda** — register-resident tridiagonal solver (H100/H200); CUDA Fortran + **CUDA C++**. github.com/k-kiha/PaScaL_TDMAcuda
- **cuda-sparse-solver-extract** — GPU sparse `Ax=b` (BiCGStab / ILU / AMG) on **cuSPARSE / cuBLAS / AmgX** in **CUDA C++**, with a C ABI for a legacy Fortran reactor code. github.com/k-kiha/cuda-sparse-solver-extract
- **PaScaL_TDMA / PaScaL_TCS** — open-source parallel & scalable solver libraries (MPMC-Lab). github.com/MPMC-Lab

EDUCATION

- **Ph.D., Computational Science and Engineering** — Yonsei University, 2020
- **B.S., Astronomy** — Yonsei University, 2014